

HAO JIANG

+1 (281)-844-6606 [◇ hj37@rice.edu](mailto:hj37@rice.edu) [◇ @JackyJiang0410](https://twitter.com/JackyJiang0410) 

EDUCATION

The University of Texas at Austin

Ph.D. in Electrical and Computer Engineering (ECE)

Human Signals Lab, Advisor: [Dr. Edison Thomaz](#)

Research Focus: Wearable Devices, Sensor Systems, and Applications in Health & Sports

Cockrell School of Engineering Fellowship Award

Austin, TX

Aug. 2025 – Present

Rice University

B.S. in Computer Science; B.A. in Sport Analytics

Major GPA: 3.82/4.00 **Distinction in Research and Creative Work**

Houston, TX

Aug. 2021 – May 2025

RESEARCH EXPERIENCE

Open-World Continual Human Activity Recognition

Austin, TX

Graduate Research Assistant, Advisor: [Dr. Edison Thomaz](#), Human Signals Lab, The University of Texas at Austin

Aug. 2025 – Present

- Developed an open-world continual human activity recognition framework that identifies known activities, detects previously unseen activities, and incrementally discovers novel activity classes from streaming wearable sensor data.
- Designed a dual-representation architecture that separates activity recognition from novel-class discovery, enabling robust recognition of known classes while maintaining discriminative representations for emerging activities.
- Developed a prototype-based recognition and novelty-detection pipeline with an unknown-sample buffer and periodic clustering for discovering recurring unseen activity patterns.
- Evaluated the system across multiple wearable activity-recognition datasets and user-generalization settings, studying recognition accuracy, novel-class discovery, and catastrophic forgetting under continuous learning.
- Published this work at the [4th International Workshop on Human-Centered Sensing, Modeling, and Intelligent Systems \(HumanSys 2026\)](#).

SeamFit: Smart Clothing for Automatic Exercise Logging

Ithaca, NY

Research Assistant, Advisor: [Dr. Cheng Zhang](#), SciFi Lab, Cornell University

Apr. 2024 – May 2025

- Designed and developed a wearable sensing system using sensors embedded in clothing seams to collect multi-channel motion signals during exercise.
- Developed deep learning models combining temporal and spatial sensor representations for exercise recognition, achieving over 92% accuracy in cross-user evaluations.
- Applied Dynamic Time Warping (DTW) to repetition counting from recognized exercise signals, achieving less than 10% counting error.
- Integrated sensing, exercise recognition, and repetition counting into an end-to-end exercise logging pipeline for practical smart-clothing applications.
- Contributed to [SeamFit: Towards Practical Smart Clothing for Automatic Exercise Logging](#), published in *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)*, 2025.

Multi-Modal Wearable Sensing for Exercise Recognition

Ithaca, NY

Research Assistant, Advisor: [Dr. Cheng Zhang](#), SciFi Lab, Cornell University

May 2024 – May 2025

- Developed a multimodal wearable sensing system integrating acoustic sensors in glasses with a wrist-worn IMU for exercise monitoring.
- Built data-processing pipelines to synchronize acoustic and inertial sensor streams and improve signal quality for multimodal learning.
- Implemented Transformer-based models for exercise recognition, achieving over 85% accuracy in cross-user evaluations.
- Evaluated multimodal fusion strategies for real-time exercise recognition using complementary acoustic and motion sensing signals.

GoalNet: A GNN-Based Soccer Player Evaluation System

Lead Researcher, Advisor: [Dr. Anastasios Kyrillidis](#), Rice University

- Designed a Graph Neural Network (GNN) framework for modeling soccer match events and evaluating individual player contributions from spatial and temporal interactions.
- Represented matches as dynamic graphs in which players are modeled as nodes and event-level interactions as edges, incorporating graph structure and centrality information into player evaluation.
- Developed methods to attribute Expected Threat (xT) changes across players and actions, enabling evaluation of contributions beyond traditional attacking statistics.
- Analyzed the framework’s ability to identify undervalued and tactically important players, including defensive and transitional roles that are poorly captured by conventional metrics.
- Published the work as the [arXiv preprint *Unveiling Hidden Pivotal Players with GoalNet: A GNN-Based Soccer Player Evaluation System*](#) in 2025.

Houston, TX

Aug. 2024 – May 2025

Context-Aware Citation Recommendation with BERT and GCN

Research Assistant, Advisor: [Dr. Arko Barman](#), D2K Lab, Rice University

- Constructed a citation-recommendation dataset from PeerRead papers by converting scientific documents into structured HTML and extracting citation contexts.
- Analyzed citation contexts and document structure to identify features relevant to context-aware scientific citation recommendation.
- Developed a recommendation pipeline combining BERT-based contextual representations with Graph Convolutional Networks (GCNs) over citation relationships.
- Investigated the use of semantic context and citation-network structure to improve the relevance of recommended academic references.

Houston, TX

Jan. 2024 – Jun. 2024

INTERNSHIP EXPERIENCE

D.C. United | Software Development Engineer Intern

May 2024 – Aug. 2024

- Contributed to the development of a comprehensive scouting app, increasing scouted players by 20%.
- Engineered a system for automated match report generation, cutting post-match analysis time by 40%.
- Created a system for matching player names and data across soccer-related websites, ensuring data consistency.
- Provided insights into emerging trends and strategies, influencing tactical decision-making for upcoming matches.

Petkeley AI Innovations | Machine Learning Engineer Intern

Dec. 2023 – Jun. 2024

- Engineered machine learning algorithms to match users for social interactions based on their virtual pet profiles, increasing daily active users by 20%.
- Developed recommendation algorithms to curate personalized user feeds.
- Improved the user matching algorithm, resulting in a 30% increase in successful connections.

ArtConnect | Machine Learning Engineer Intern

Nov. 2023 – May 2024

- Engineered a user-content recommendation system, resulting in a 35% rise in repeated platform visits.
- Implemented a Two Towers model architecture, improving content recommendation accuracy by 22%.
- Utilized large language models for feature extraction, enhancing the platform’s ability to predict artist needs.

SKILLS

Programming: Python, Java, HTML/CSS, JavaScript, SQL, R

Developer Tools: VS Code, Eclipse, Google Cloud Platform, Heroku, Cloudinary, Jupyter Notebook, Docker

Technologies/Frameworks: Linux, React, GitHub, JUnit, scikit-learn, Node.js, MongoDB, TensorFlow, PyTorch

Data Science: Machine Learning, Data Preprocessing, Feature Engineering, NumPy, Pandas, Matplotlib, seaborn, Hugging Face